

Xueyan Zhang

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RESEARCH INTERESTS

- **Efficient LLMs, end-to-end**: from training (parameter-efficient fine-tuning, optimizers, model merging) to systems (compiler & runtime optimization, operator fusion, inference engines, and hardware-aware codegen).
- Also drawn to **agentic systems** (multi-agent collaboration, self-evolving agents, agent-harness engineering), **post-training** (SFT + RL + self-distillation), and **robotics**.

EDUCATION

- **University of Waterloo**, *Master of Math in Computer Science* Waterloo, Ontario
Research Stream in Software Engineering; Avg. 93/100 *Jan. 2021 – Dec. 2022*
Graduate Excellence Award in Computer Science; International Master’s Award of Excellence
- **McMaster University**, *Bachelor of Engineering in Mechatronics Co-op (with Distinction)* Hamilton, Ontario
GPA: 11.6/12 *Sept. 2015 – Apr. 2020*
Dean’s Honour List; Senate Scholarship; Woo Family International Scholarship

PUBLICATIONS

- **UORA: Uniform Orthogonal Reinitialization Adaptation in PEFT** *ACL 2025*
 - **Co-first author**. A novel interpolation-based reinitialization mechanism for parameter-efficient fine-tuning (**PEFT**), reducing trainable parameters by up to **15x** vs. LoRA while staying competitive across NLU, NLG, instruction-tuning, and ViT.
- **Tiny Budgets, Big Gains: Parameter Placement Strategy in PEFT** *EMNLP 2025*
 - Introduced **FoRA-UA**, a sparse Fourier low-rank method with a universal shared adaptor, reaching SOTA with only 1–5% of LoRA’s parameters.
- **Large Language Model is not a (Multilingual) Compositional Relation Reasoner** *COLM 2024*
 - **Co-first author**. Created and open-sourced the **MCR Benchmark**, a multilingual evaluation of LLM compositional reasoning across six relation categories in five languages.
- **On the Caching Schemes to Speed Up Program Reduction** *TOSEM 2024*
 - Master’s research. Proposed Refreshable Compact Caching (**RCC**), cutting **64%** redundant queries, accelerating reduction by up to **43%**, and shrinking memory **99.97%** (100+ GB → 6 MB).
- **Ethical and Robust Large Language Models** *WWW 2026 Workshop*
 - Studied the ethical risks of 3 LLM reasoning frameworks (CoT, Self-Reflection, Multi-Agent Debate) across 6 LLMs and 4 languages, quantifying a reasoning-vs-safety trade-off — Multi-Agent Debate dropped reasoning safety by **34.4%** on average.

PREPRINTS

- **How Much Operator Fusion Is Left? A Compiler-Side Bound & Learned Policy** *AAAI 2027*
 - **First author**. Derived a compiler-side upper bound on remaining operator-fusion gains and trained a learned graph-level fusion policy (SFT+GRPO) for PyTorch Inductor, reaching avg. **15%** end-to-end speedup (up to **3x**) over torch.compile. *Under review*.
- **HR-Merge: Model Merging with Headroom-Weighted Residual Refinement** *AAAI 2027*
 - **Co-first author**. A model-merging method using headroom-weighted residual refinement to combine multiple fine-tuned models. *Under review*.
- **TARE-Muon: Timescale-Aware Row-wise Energy Redistribution** *AAAI 2027*
 - A timescale-aware, row-wise energy-redistribution optimizer (Muon family) using long- and short-term momentum. *Under review*.
- **ATPO: Adaptive Token Aggregation for Policy Optimization** *AAAI 2027*
 - Adaptive token aggregation for policy optimization in RL fine-tuning of LLMs. *Under review*.
- **IntentEval: Do LLMs Answer What Users Actually Mean?** *EMNLP 2026*
 - A benchmark for whether LLMs answer users’ true intent when questions are underspecified, vague, or carry false presuppositions. *Under review*.

- **Syntactic Prediction through Reinforcement Learning** EMNLP 2026
 - A reinforcement-learning approach to syntactic prediction in language models. *Under review.*
- **Can Language Model Understand Word Semantics as A Chatbot?** Preprint
 - **Co-first author.** Quantified internal-vs-external knowledge mismatch for word semantics across Encoder, Decoder, and Encoder-Decoder LMs; probing internal states recovers more accurate semantics than querying outputs. *arXiv preprint.*

EXPERIENCE

- **Heterogeneous Compiler Lab, Huawei Technology Canada** Toronto, Ontario
Senior NPU Compiler Engineer Sept. 2022 – June 2026
 - **Ascend NPU Compiler Optimization for LLMs:**
 - * Spearheaded compiler optimizations in LLVM for featured networks, like **Flash Attention** and RMS norm, achieving **25%** improvement in operator performance and a 10% reduction in compiled binary size.
 - * Served as the technical focal point for a small-sized team, leading communication and aligning technical progress with international counterparts.
 - * Designed a suite of custom compiler passes: enhanced recognition and transformation of zero-overhead hardware loops, plus ahead-of-time constant propagation, prefetching, custom loop fusion, and peephole optimizations.
 - **Unified Test Case Reduction Framework:**
 - * Architected an award-winning test case reduction framework that industrializes my **program-reduction research**: unified 5 open-source algorithms and pioneered a novel, LLM-based approach to auto-generate test properties.
 - * This service is now used by **200+** developers and has boosted debugging efficiency by 50%, automatically minimizing a 10k-line test case to its 100-line root cause in under 30 minutes.
 - **AI-powered Code Review Service:**
 - * Built an AI-powered code review agent on **Jenkins**, integrating **Deepseek** as the LLM backend and **OpenCode** as the agent framework, with multi-agent collaboration for automated merge request analysis.
 - * Configured GitLab CI pipelines and provided glab CLI tooling for MR view, diff, and API operations, enabling fully automated review feedback on every merge request.
 - * Deployed Deepseek V4 Flash with 1M context window via **vllm** on two-node NVIDIA GB10.
 - **Data Movement Interface Re-architecture:** Redesigned data movement interface, introducing LLVM intrinsics and MIR pseudos that eliminated error-prone code and simplified analysis for downstream passes.
 - **General Matrix Multiplication in MLIR:** Engineered a vectorization feature in MLIR to pattern match, transform, and optimize user codes to fully utilize the matrix computing unit. Supported double buffering.
 - **Student Intern Mentorship:** Mentored multiple software engineering interns on compiler projects, leading to the successful integration of their features into the production codebase and a final results presentation to headquarters.
- **UWaterloo CS136, Instructional Apprentice** Waterloo, Ontario
Clarified algorithms and data-structure concepts in weekly office hours. Sept. 2021 – May 2022
- **UWaterloo SWAG Lab, Undergraduate Research Assistant** Waterloo, Ontario
Engineered research benchmarking automation with tracking, quantifying, and visualizing. Sept. 2020 – Dec. 2020

PROJECTS

- **OpenMarkdown.dev, Local-first Markdown editor with agent co-editing** Jul. 2026
 - Built a local-first desktop editor with a **Rust** core that exposes a **Model Context Protocol (MCP)** interface for real-time AI-agent co-editing; **reached #5 on Product Hunt** at launch.
 - Designed a conflict-resolution scheme that merges concurrent human and agent edits without data loss.
- **OpenTikZ.org, Open-source TikZ library for AI-assisted academic diagrams** May 2026
 - Designed and shipped a community TikZ resource library, the “Flaticon for academic TikZ”, delivering **100+** reusable assets for system, neural-network, and pipeline figures; **240+** GitHub stars.
 - Shipped a **Claude Code** skill, enabling an AI agent to discover, edit, and recompile figures from natural-language requests so researchers produce publication-ready diagrams without hand-writing TikZ.
- **iMarkAi.com, Agentic bookmark with semantic search & web interactions** Nov. 2025
 - Designed, implemented, and published a commercial-grade bookmark extension on Chrome Web Store.
 - Architected an intelligent search platform using **Pinecone** vector database and Google Gemini AI, achieving sub-second semantic retrieval across unstructured web data with 95%+ precision.
 - Implemented end-to-end **Stripe** billing for subscriptions and usage-based payments.
- **Tarot: Starry Guidance, End-to-end mobile app development** Sept. 2025
 - Led the end-to-end, full-stack development on the WeChat mini-program platform; launched within 1 month.
 - Utilized web scraping for data; implemented HTML/CSS for UI, a cloud database, and an integrated LLM for personalized interpretations of drawn cards.
 - Conducted market research and competitor analysis; **reached 500+ users in first 3 months.**

TECHNICAL SKILLS

- **Programming:** C/C++,  Python, Bash,  Java/Kotlin,  Unix/Linux, **git** Git
- **Toolbox:** Huggingface, Pytorch, LLVM, MLIR, Bazel,  Docker, Claude Code/API/SDK, NVIDIA Spark

SERVICE

- Reviewer, ICLR 2025; AAAI 2027; ACL ARR 2026
- Delivered Prompt Engineering, Reinforcement Learning, and Fine-tuning Crash Courses
- PC hardware enthusiast for ultra-compact ITX builds